

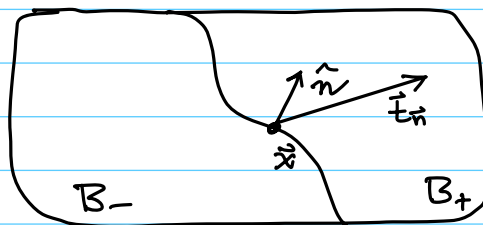
Continuum Mechanics

class #14

9/22/2026

Recap

let a body $B = B_- \cup B_+$, where $B_- \cap B_+ = \emptyset$ and the dividing surface Γ is sufficiently regular.



The force per unit area exerted by the positive side, B_+ , on the negative side, B_- , is called

$$\vec{t}_{\hat{n}}(\cdot) : \Gamma \rightarrow \mathcal{V}.$$

It satisfies

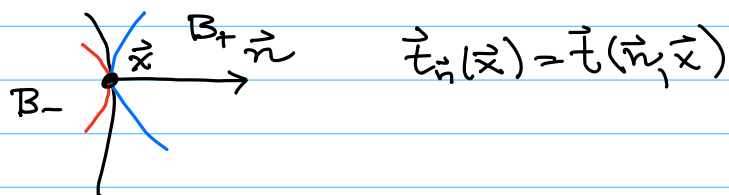
$$\vec{t}_{\hat{n}}(\vec{x}) = \vec{t}(\hat{n}(\vec{x}), \vec{x})$$

for some function

$$\vec{t} : N \times \mathbb{E}^3 \rightarrow \mathcal{V},$$

which is known as Cauchy's postulate. In other words, the traction only depend up on the point of application and the normal vector.

The tractions do not depend upon curvatures of Γ , for example.



The principle of action - reaction requires

$$\vec{t}(\vec{n}, \vec{x}) = -\vec{t}(-\vec{n}, \vec{x}).$$

In other words the traction exerted by B_- on B_+ is equal and opposite to the traction exerted by B_+ on B_- .

This was derived by a more basic assumption of action - reaction, namely, that $\vec{t}: N \times \mathbb{E}^3 \rightarrow V$ satisfies

$$(3.1) \quad \frac{1}{\text{area}(\partial\Omega)} \int_{\partial\Omega} \vec{t}(\vec{n}(\vec{x}), \vec{x}) dA_x \rightarrow 0$$

as

$$\text{Vol}(\Omega) \rightarrow 0.$$

We will come back to this discussion when we talk about the balance of linear momentum.

As another consequence of (3.1), we have

$$\vec{t}(\vec{n}, \vec{x}) = S(\vec{x})\hat{n},$$

for some second-order tensor $S: \mathbb{E}^3 \rightarrow V^2$ called the Cauchy stress tensor.

We finally showed that at equilibrium, when there are no total resultant forces or torques in the body, the conditions for equilibrium may be expressed locally as

$$(3.8) \quad (\nabla \cdot \mathbf{S})(\vec{x}) + \rho(\vec{x}) \vec{b}(\vec{x}) = 0 \quad \forall \vec{x} \in B.$$

$$\mathbf{S}^T(\vec{x}) = \mathbf{S}(\vec{x})$$

Simple States of Stress

Definition Suppose that

$$\mathbf{S}: B \rightarrow \mathcal{V}^2$$

is the stress tensor for a body occupying $B \subseteq \mathbb{E}^3$.
If at $\vec{x} \in B$

$$\mathbf{S}(\vec{x}) = -\pi \mathbf{I}$$

where $\pi \in \mathbb{R}$, we say that a spherical state of stress exists at $\vec{x}$. //

For such a stress state,

$$\vec{t}(\vec{n}, \vec{x}) = -\pi \vec{n}$$

for any $\vec{n}$. In other words the traction on any surface with normal $\vec{n}$ going through the point $\vec{x}$ is always

$$\vec{t} = -\pi \hat{n}.$$

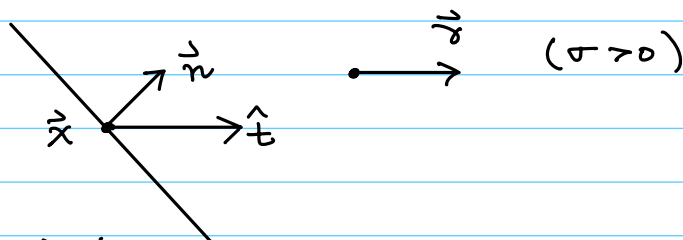
Definition: The state of stress is called uniaxial at $\vec{x} \in B$ iff there is a unit vector $\vec{s}$ such that

$$S(\vec{x}) = \sigma \vec{s} \otimes \vec{s}.$$

Such a state is called pure tension if $\sigma > 0$ and pure compression if $\sigma < 0$. //

For such a state the traction on any surface with normal $\vec{n}$ at $\vec{x}$ is

$$\vec{t} = S\hat{n} = (\vec{s} \cdot \vec{n}) \sigma \vec{s}$$



If $\vec{s} \perp \vec{n}$, i.e. $\vec{s} \cdot \vec{n} = 0$, then

$$\vec{t} = \vec{0}$$

Definition: If at a point $\vec{x} \in B$ there is a pair of orthogonal unit vectors $\vec{s}, \vec{n} \in N$, $\vec{s} \cdot \vec{n} = 0$, and a scalar γ such that

$$S(\vec{x}) = \gamma (\vec{s} \otimes \vec{n} + \vec{n} \otimes \vec{s}),$$

we say that a state of pure shear exists at $\vec{x}$. //

For such a state, the traction on any surface with normal $\vec{n}$ at $\vec{x}$ is given by

$$\vec{t} = S\vec{n} = (\vec{\eta} \cdot \vec{n})\gamma\vec{\delta} + (\vec{\delta} \cdot \vec{n})\gamma\vec{\eta}.$$

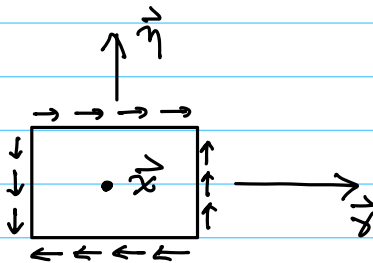
When $\vec{n} = \vec{\eta}$,

$$\vec{t} = \gamma\vec{\delta}.$$

When $\vec{n} = \vec{\delta}$,

$$\vec{t} = \gamma\vec{\eta}.$$

For example,



the surface tractions on the box surrounding $\vec{x}$ are parallel to the bounding surfaces and oriented as indicated.

The tractions on the surfaces of the box whose normals are out of or into the plane are zero.

Example: Suppose that

$$S = \gamma(\vec{\delta} \otimes \vec{n} + \vec{\eta} \otimes \vec{\delta}).$$

Assume that our frame is

$$\mathcal{F} = \{\vec{\delta}, \vec{\eta}, \vec{e}_3\} \quad \vec{e}_3 = \vec{\delta} \times \vec{\eta}$$

Then

$$[S] = \begin{bmatrix} 0 & \tau & 0 \\ \tau & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$S_{11} = \vec{\delta} \cdot (S\vec{\delta})$$

$$= \vec{\delta} \cdot \tau \vec{\eta}$$

$$= 0$$

$$S_{12} = \vec{\delta} \cdot S\vec{\eta}$$

$$= \vec{\delta} \cdot (\tau \vec{\delta})$$

$$= \tau$$

$$S_{13} = \vec{\delta} \cdot S\vec{e}_3$$

$$= 0.$$

You can easily finish the calculation/confirmation.

Definition: If at a point $\vec{x} \in B$ there exists a pair $\vec{\delta}, \vec{\eta} \in \mathcal{V}$, $\vec{\delta} \cdot \vec{\eta} = 0$, such that in the frame $\mathcal{F} = \{\vec{\delta}, \vec{\eta}, \vec{\delta} \times \vec{\eta}\}$

$$[S(\vec{x})] = \begin{bmatrix} S_{11} & S_{12} & 0 \\ S_{21} & S_{22} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

we say that there is a state of plane stress exists at $\vec{x}$.

Clearly pure shear is a state of plane stress.

Principal, Normal, and Shear Stresses

Definition: let $S: B \rightarrow \mathcal{V}^2$ be the stress state of a body in B in equilibrium. The eigenvalues of $S(\vec{x})$,

$$\sigma_1(\vec{x}), \sigma_2(\vec{x}), \sigma_3(\vec{x}) \in \mathbb{R}$$

are called the principal stresses. The three associated orthonormal eigenvectors

$$\vec{e}_1(\vec{x}), \vec{e}_2(\vec{x}), \vec{e}_3(\vec{x})$$

are called the principal stress directions at $\vec{x}$. Thus,

$$S \vec{e}_i = \sigma_i \vec{e}_i \quad (\text{no sum}) //$$

For a surface with normal $\vec{e}_i$

$$\vec{t} = S \vec{e}_i = \sigma_i \vec{e}_i, \quad (\text{no sum})$$

ie, the traction is in the direction of $\vec{e}_i$.

Definition: let $S: B \rightarrow \mathcal{V}^2$ be the stress state of a body in equilibrium. Assume Γ is a surface with normal $\hat{n}$ at $\vec{x}$



The traction on Γ at $\vec{x}$ is

$$\vec{T}(\vec{x}) = S(\vec{x})\vec{n}.$$

The vector

$$\vec{T}_n = (\vec{T} \cdot \vec{n})\vec{n}$$

is called the normal traction. The vector

$$\vec{T}_s = \vec{T} - \vec{T}_n$$

is called the shear traction. Clearly

$$\vec{T}_n + \vec{T}_s = \vec{T}.$$

We call $\sigma_n := |\vec{T}_n|$ the normal stress and $\sigma_s := |\vec{T}_s|$ the shear stress. //

Example: Suppose in the context of the last definition surface unit normal is

$$\vec{n} = \vec{e},$$

where $\vec{e}$ is one of the principal stress directions associated to the principal stress, σ , i.e.

$$\vec{T} = S\vec{e} = \sigma\vec{e}$$

Then,

$$\vec{T}_n = (\vec{T} \cdot \vec{e})\vec{e} = (\sigma\vec{e} \cdot \vec{e})\vec{e} = \sigma\vec{e} = \vec{T}$$

It must be that

$$\vec{t}_s = \vec{0}$$

The normal stress is

$$\sigma_n = |\sigma|$$

and the shear stress is zero.

Maximal Normal and Shear Stresses

Result (3.6): Assume that, at a point $\vec{x} \in B$, the 3 principal stresses of $S(\vec{x})$ are distinct and ordered such that

$$(3.12) \quad \sigma_1 > \sigma_2 > \sigma_3.$$

Then

(1) The maximum value of σ_n is

$$|\sigma_k| = \max \{|\sigma_1|, |\sigma_3|\}.$$

Here $k=1$ or $k=3$. This value is achieved for the direction $\vec{n} = \pm \vec{e}_k$.

(2) The maximum value of σ_s is

$$\frac{1}{2} |\sigma_1 - \sigma_3|$$

This value is achieved for the pairs

$$\vec{n} = \pm \frac{1}{\sqrt{2}} (\vec{e}_1 + \vec{e}_3) \quad \pm \frac{1}{\sqrt{2}} (\vec{e}_1 - \vec{e}_3)$$

Proof: The max of σ_n is proven in Exercise 3.17.
The max for σ_s is proven in Exercise 3.18. ///

Spherical and Deviatoric Stress Tensors

Proposition: Let $S: B \rightarrow V^2$ be the stress tensor of a body (typically, at equilibrium). Then there are two tensors $S_s, S_D \in V^2$ such that

$$\text{tr}(S_s) = \text{tr}(S)$$

and

$$\text{tr}(S_D) = 0,$$

where S_s is spherical, i.e.,

$$S_s = -pI.$$

Proof: Set

$$p = -\frac{1}{3} \text{tr}(S).$$

and

$$S_s = -pI.$$

Then

$$\text{tr}(S_s) = -p \text{tr}(I)$$

$$= \frac{1}{3} \text{tr}(S) \cdot 3$$

$$= \text{tr}(S).$$

Now set

$$S_D = S - S_S$$

Then

$$\text{tr}(S_D) = \text{tr}(S) - \text{tr}(S_S)$$

$$= 0. \quad ///$$

Remark: At equilibrium $S = S^T$. The 3 eigenvalues exist and

$$\text{tr}(S) = \sigma_1 + \sigma_2 + \sigma_3$$

Hence

$$p = -\frac{1}{3}(\sigma_1 + \sigma_2 + \sigma_3).$$

Here p is called the pressure.

Definition: The tensor S_S is called the spherical stress tensor and S_D is called the deviatoric stress tensor.

Roughly speaking, the spherical stress is that part of the stress that tends to change the volume of a body, but not its shape.

The sign convention for pressure is such that $p > 0$ corresponds to a state of compression.

Roughly speaking, the deviatoric stress is that part of the overall stress that tends to change the shape of a body while producing little or no change in volume.

Definition: let S_D be the deviatoric part of the stress tensor S . Then,

$$J_2(S) := -I_2(S_D) = \frac{1}{2} S_D : S_D$$

and

$$J_3(S) := I_3(S_D).$$